K-Means Clustering – Solved Example

- Suppose that the data mining task is to cluster points into three clusters,
- where the points are
- *A*1(2, 10), *A*2(2, 5), *A*3(8, 4), *B*1(5, 8), *B*2(7, 5), *B*3(6, 4), *C*1(1, 2), *C*2(4, 9).
- The distance function is Euclidean distance.
- Suppose initially we assign *A*1, *B*1, and *C*1 as the center of each cluster, respectively.

Point	Cord.	dist c_1	dist c_2	dist c_3	cluster
A_1	(2, 10)	0	3.6	8.06	c_1
A_2	(2, 5)	5	4.24	3.16	c_3
A_3	(8, 4)	8.48	5	7.28	c_2
B_1	(5, 8)	3.6	0	—	c_2
B_2	(7, 5)	7.07	3.6	6.7	c_2
B_3	(6, 4)	7.2	4.12	5.38	c_2
C_1	(1, 2)	—	—	0	c_3
C_2	(4, 9)	2.2	1.4	7.6	c_2

$$c_1 = (2, 10)$$
$$c_2 = (5, 8)$$
$$c_3 = (1, 2)$$

$$c_1 = (2, 10)$$
$$c_2 = (8, 4), (5, 8), (7, 5), (6, 4), (4, 9)$$
$$c_3 = (1, 2), (2, 5)$$

$$c_1 = (2, 10)$$
$$c_2 = (6, 6)$$
$$c_3 = (1.5, 3.5)$$